破解紀錄整理:

1. **初始雜湊破解**：hash1.txt

- **工具與指令**：使用 hashcat 搭配 NTLM 模式（-m 1000），並載入 best66.rule 變化規則。
- **破解過程**：先執行 rm -f ...potfile 清理舊紀錄以確保現場顯示，再載入 rockyou.txt 字典進行大規模比對。
- **終端輸出**：系統成功匹配雜湊值，狀態顯示為 Cracked。
- **原始明文**：最終破獲的原始密碼內容為 thesunsh1n3。

```
┌──(kali㉿kali)-[~/Downloads/redo_final/hw]
└─$ cd ~/Downloads/redo_final/hw
rm -f ~/.local/share/hashcat/hashcat.potfile

┌──(kali㉿kali)-[~/Downloads/redo_final/hw]
└─$ echo "04eebfb5611c3a37b97701c059d5e7c0" > hash1_pure.txt

┌──(kali㉿kali)-[~/Downloads/redo_final/hw]
└─$ hashcat -m 1000 hash1_pure.txt /usr/share/wordlists/rockyou.txt -r /usr/share/hashcat/rules/best66.rule --force
hashcat (v7.1.2) starting

You have enabled --force to bypass dangerous warnings and errors!
This can hide serious problems and should only be done when debugging.
Do not report hashcat issues encountered when using --force.

OpenCL API (OpenCL 3.0 PoCL 6.0+debian  Linux, None+Asserts, RELOC, SPIR-V, LLVM 18.1.8, SLEEF, DISTRO, POCL_DEBUG) - Platform #1 [The pocl project]
```

```
Host memory allocated for this attack: 513 MB (2499 MB free)

Dictionary cache hit:
* Filename..: /usr/share/wordlists/rockyou.txt
* Passwords.: 14344385
* Bytes.....: 139921507
* Keyspace..: 946729410

04eebfb5611c3a37b97701c059d5e7c0:thesunsh1n3

Session..........: hashcat
Status...........: Cracked
Hash.Mode........: 1000 (NTLM)
Hash.Target......: 04eebfb5611c3a37b97701c059d5e7c0
Time.Started.....: Sun Apr 12 03:39:32 2026, (0 secs)
Time.Estimated...: Sun Apr 12 03:39:32 2026, (0 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.......: File (/usr/share/wordlists/rockyou.txt)
Guess.Mod........: Rules (/usr/share/hashcat/rules/best66.rule)
Guess.Queue......: 1/1 (100.00%)
Speed.#01........: 31155.5 kH/s (3.81ms) @ Accel:1024 Loops:64 Thr:1 Vec:8
Recovered........: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.........: 17293312/946729410 (1.83%)
Rejected.........: 0/17293312 (0.00%)
Restore.Point....: 258048/14344385 (1.80%)
Restore.Sub.#01..: Salt:0 Amplifier:0-64 Iteration:0-64
Candidate.Engine.: Device Generator
Candidates.#01...: tubby12 → el92
Hardware.Mon.#01.: Util: 54%

Started: Sun Apr 12 03:39:31 2026
Stopped: Sun Apr 12 03:39:34 2026
```

2. **權杖金鑰破解**：token.txt (JWT)

- **工具與指令**：使用 hashcat 的 JWT 模式 (-m 16500)，目標是破解簽章段的密鑰。
- **破解過程**：手動校正 token 內容後，搭配由 cewl 從淡大官網爬取的 tku.wordlist 專用字典進行精確爆破。
- **終端輸出**：成功識別加密段分隔符並完成匹配，狀態標示為 Cracked。
- **原始明文**：破獲用於簽署 token 的 Secret Key 為 intercollegiate。

```
┌──(kali㉿kali)-[~/Downloads/redo_final/hw]
└─$ cd ~/Downloads/redo_final/hw

┌──(kali㉿kali)-[~/Downloads/redo_final/hw]
└─$ echo "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJOZXh0LVN0ZXAiOiJjZXdsIGh0dHBzOi8vZW5nbGlzaC50a3UuZWR1LnR3Ly
AtZCAyIC1tIDggLS1sb3dlcmNhc2UgLXcgdGt1LndvcmRsaXN0IiwiaWF0IjoxNzQwMzc0Njg1fQ.zZzR3owkuGQ8P1NZGb1QNUAsoQjFwqRD
Zk3leuSa1e8" > hash2.txt

┌──(kali㉿kali)-[~/Downloads/redo_final/hw]
└─$ rm -f ~/.local/share/hashcat/hashcat.potfile

┌──(kali㉿kali)-[~/Downloads/redo_final/hw]
└─$ hashcat -m 16500 hash2.txt tku.wordlist --force
hashcat (v7.1.2) starting

You have enabled --force to bypass dangerous warnings and errors!
This can hide serious problems and should only be done when debugging.
Do not report hashcat issues encountered when using --force.

OpenCL API (OpenCL 3.0 PoCL 6.0+debian  Linux, None+Asserts, RELOC, SPIR-V, LLVM 18.1.8, SLEEF, DISTRO, POCL_
DEBUG) - Platform #1 [The pocl project]
==================================================================================================
* Device #01: cpu-haswell-13th Gen Intel(R) Core(TM) i9-13900H, 1456/2912 MB (512 MB allocatable), 4MCU
```

```
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJOZXh0LVN0ZXAiOiJjZXdsIGh0dHBzOi8vZW5nbGlzaC50a3UuZWR1LnR3LyAtZCAyIC1t
IDggLS1sb3dlcmNhc2UgLXcgdGt1LndvcmRsaXN0IiwiaWF0IjoxNzQwMzc0Njg1fQ.zZzR3owkuGQ8P1NZGb1QNUAsoQjFwqRDZk3leuSa1e
8:intercollegiate

Session..........: hashcat
Status...........: Cracked
Hash.Mode........: 16500 (JWT (JSON Web Token))
Hash.Target......: eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJOZXh0LVN0ZX...uSa1e8
Time.Started.....: Sun Apr 12 03:41:55 2026, (0 secs)
Time.Estimated...: Sun Apr 12 03:41:55 2026, (0 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.......: File (tku.wordlist)
Guess.Queue......: 1/1 (100.00%)
Speed.#01........:   795.7 kH/s (0.56ms) @ Accel:1024 Loops:1 Thr:1 Vec:8
Recovered........: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.........: 1202/1202 (100.00%)
Rejected.........: 0/1202 (0.00%)
Restore.Point....: 0/1202 (0.00%)
Restore.Sub.#01...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#01...: department → workshops
Hardware.Mon.#01.: Util: 27%

Started: Sun Apr 12 03:41:54 2026
Stopped: Sun Apr 12 03:41:57 2026
```

3. **資料庫雜湊破解：hw.db (PYC 帳號)**

- **工具與指令**：使用 hashcat 搭配 SHA-512 模式 (-m 1700)，針對資料庫提取出的加密字串。
- **破解過程**：解開 hw.7z 獲取資料庫後，將正確雜湊存入 password.txt，並在字典中補齊特定關鍵字後執行最終驗證。
- **終端輸出**：破解效率極高，瞬間於畫面上噴出正確的明文結果，狀態顯示為 Cracked。
- **原始明文**：最終破獲該帳號的原始密碼為中文內容：**淡水巨無霸冰淇淋**。

```
┌──(kali㉿kali)-[~/Downloads/redo_final/hw]
└─$ rm -f ~/.local/share/hashcat/hashcat.potfile

┌──(kali㉿kali)-[~/Downloads/redo_final/hw]
└─$ hashcat -m 1700 password.txt tku.wordlist --force
hashcat (v7.1.2) starting

You have enabled --force to bypass dangerous warnings and errors!
This can hide serious problems and should only be done when debugging.
Do not report hashcat issues encountered when using --force.

OpenCL API (OpenCL 3.0 PoCL 6.0+debian  Linux, None+Asserts, RELOC, SPIR-V, LLVM 18.1.8, SLEEF, DISTRO, POCL_
DEBUG) - Platform #1 [The pocl project]
==================================================================================================
* Device #01: cpu-haswell-13th Gen Intel(R) Core(TM) i9-13900H, 1456/2912 MB (512 MB allocatable), 4MCU

Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 256

Hashes: 1 digests; 1 unique digests, 1 unique salts
Bitmaps: 16 bits, 65536 entries, 0x0000ffff mask, 262144 bytes, 5/13 rotates
Rules: 1
```

```
The wordlist or mask that you are using is too small.
This means that hashcat cannot use the full parallel power of your device(s).
Hashcat is expecting at least 4096 base words but only got 29.4% of that.
Unless you supply more work, your cracking speed will drop.
For tips on supplying more work, see: https://hashcat.net/faq/morework

Approaching final keyspace - workload adjusted.

4d5d085ef162d75520cc048b7d6db6513039044d28d15247128dbd610d33f7c32943e914d6a35329f4f536281ab11dbad29378108c655
82ce7193b7bc52d885b:淡水巨無霸冰淇淋

Session..........: hashcat
Status...........: Cracked
Hash.Mode........: 1700 (SHA2-512)
Hash.Target......: 4d5d085ef162d75520cc048b7d6db6513039044d28d15247128 ... 2d885b
Time.Started.....: Sun Apr 12 03:57:20 2026, (0 secs)
Time.Estimated...: Sun Apr 12 03:57:20 2026, (0 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.......: File (tku.wordlist)
Guess.Queue......: 1/1 (100.00%)
Speed.#01........:  1316.0 kH/s (0.14ms) @ Accel:1024 Loops:1 Thr:1 Vec:4
Recovered........: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.........: 1203/1203 (100.00%)
Rejected.........: 0/1203 (0.00%)
Restore.Point....: 0/1203 (0.00%)
Restore.Sub.#01..: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#01...: department → 淡水巨無霸冰淇淋
Hardware.Mon.#01.: Util: 26%

Started: Sun Apr 12 03:57:20 2026
Stopped: Sun Apr 12 03:57:22 2026
```

補充嘗試歷程與排錯方法

- **環境修復與檔案清理嘗試：**
  - **資料庫壞軌重置：**曾遇到 hw.db 檔案大小顯示為 0 bytes 的損壞情況，導致 sqlite3 無法執行查詢。我嘗試了「徹底刪除目錄並重新解壓縮」的方法，並在 7z 覆蓋提示中選擇 A (Always replace)，成功復原了 8KB 的完整資料庫檔案。
  - **Hashcat 快取管理：**發現 hashcat 會記憶先前的破解結果，導致無法在截圖中現場噴出 Cracked 字樣。我學會使用 rm -f ~/.local/share/hashcat/hashcat.potfile 來強制清除快取，這對於產出助教要求的「色彩標示畫面」至關重要。
- **指令優化與語法排錯：**
  - **自動化提取邏輯：**在提取雜湊時，我嘗試過 cut、awk 以及 sqlite3 ".dump" 等不同工具。我發現當資料格式包含特殊引號時，使用單一字元切割容易報錯；最終掌握了利用 grep 篩選關鍵字並精確提取第 4 欄位的正規化方法。
  - **解決權杖長度報錯：**曾遇到 Token length exception 錯誤，這促使我重新檢查輸入檔案 password.txt 的純淨度。我嘗試透過 echo 指令直接寫入雜湊值，避開了檔案讀取時可能產生的換行符號或隱形字元，成功解決了 hashcat 的格式判定問題。
- **字典與規則的策略調整：**
  - **多字典切換驗證：**針對不同關卡，我嘗試區分使用萬用字典 rockyou.txt 與專用字典 tku.wordlist。
  - **規則檔適配：**針對助教提出的 NTLM 破解，我嘗試將 best64.rule 更新為更強大的 best66.rule ，並驗證了在不同規則下破解速度與成功率的差異。